

APAN 5310 SQL
April 29th, 2026

Dream Homes NYC Project

Group 6

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1. Project Overview

1. Client Background

- Regional real estate agency covering the Tri-State market, focused on sales and rentals
- Manages core business entities including agents, clients, listings, and transactions
- Highly interconnected operations generate complex, multi-stage lifecycle data

2. Current Challenges

- Data is currently stored across spreadsheets and disconnected systems, leading to fragmentation and inconsistency
- Lack of a centralized system makes it difficult to:
 - Track the full lifecycle from listing to transaction
 - Analyze agent performance and client behavior
 - Measure conversion rates and operational efficiency
- As a result, management has limited visibility into key business metrics, making data-driven decision-making challenging

3. Project Objective

- Design and implement a centralized relational database system to integrate all business data into a single structured platform
- Enable:
 - Efficient data storage through normalized schema design
 - Advanced SQL-based analytics for detailed business insights
 - Dashboard reporting for executives and non-technical stakeholders

2. Business Motivation

1. Why This Project Matters

- Real estate operations involve multiple highly interconnected entities
- These entities interact across different stages of the customer lifecycle, from initial inquiry to final transaction
- Traditional spreadsheets cannot handle such complexity, leading to inefficiencies and fragmented data

2. Key Design Opportunity

- A relational database enables structured data modeling with clearly defined relationships between entities
- The listing-centered design captures the full process from listing to client engagement and transaction
- This structure supports:
 - Operational tracking (e.g., listings, appointments, transactions)
 - Analytical insights (e.g., conversion rates, demand patterns, agent performance)

3. Expected Business Impact

- Establishes a centralized and consistent data foundation, improving data quality and reliability
- Enables the company to answer complex business questions through SQL queries and integrated analysis
- Provides a scalable platform for:
 - Performance monitoring
 - Data-driven decision-making
 - Future analytics applications such as forecasting and recommendation systems

3. System Architecture

1. End-to-End Data Workflow

ETL process using Python:

- Validate schema consistency
- Standardize formats
- Ensure foreign key relationships before loading

Step 1



Data generation in structured CSV format

Step 2



Storage in PostgreSQL relational database

Step 3



Analytical queries using SQL

Step 4



Visualization via Metabase dashboards

Step 5



2. User Interaction Layers

➤ Analysts (Technical Users)

- Interact directly with PostgreSQL via SQL and pgAdmin
- Perform detailed analysis on agent performance, client engagement, and transaction trends

➤ Executives (Non-Technical Users)

- Access insights through Metabase dashboards
- Monitor KPIs such as listings, revenue, conversion rates, and market trends

3. Key Advantages

- Centralized data platform replacing fragmented spreadsheets
- Scalable and reproducible analytics through SQL and Python workflows
- Flexible access model supporting both deep analysis and high-level reporting
- Bridges the gap between technical data processing and business decision-making

4. Dataset Overview

1. Dataset Characteristics

- 1** **Source:** synthetic dataset designed to align closely with the Dream Homes NYC business scenario
- 2** **Format:** structured CSV files prepared for relational database loading
- 3** **Scale:** 13,080 rows across 16 interconnected tables, covering both operational and transactional data

3. Data Design Considerations

- Data is generated to match the schema structure, reducing heavy cleaning requirements
- Ensures consistency across foreign key relationships
- Supports a normalized, multi-table relational design

2. Core Data Coverage

Operational Data: offices, employees, agents

Customer Data: clients, client preferences

Business Activities: listings, inquiries, appointments, open houses

Transactions: property sales and rental agreement

- Covers the end-to-end real estate lifecycle

4. Business Value

- Provides a complete and structured representation of operations
- Enables:
 - operational tracking across the pipeline
 - analytical modeling for performance and demand
- Serves as a reliable foundation for SQL analysis and dashboards

5. Database Design (ERD & Logic)

1. Design Approach: Listing-Centered Model

- The database is structured around listings as the central entity, linking properties, clients, and transactions
- Separates core concepts into:
 - property (asset)
 - listing (market activity)
 - transaction (final outcome)

3. Importance

- Reflects the real-world business workflow
- Enables analysis of:
 - conversion rates
 - client engagement behavior
 - market demand patterns
 - agent performance

2. Key Concept

- **Property** → Physical real estate asset
- **Listing** → A specific listing event for that property
- **Transaction** → Completed sale or rental outcome
- Distinguishes **static entities vs. event-driven processes**

Relationship Flow

- Listing → client engagement (inquiries, appointments, open houses) → transaction
- Captures the full pipeline from supply → interaction → conversion

ERD Visualization:

https://lucid.app/lucidchart/a24bd5b7-3b4f-4cac-801a-9d881439bb8e/edit?viewport_loc=462%2C-732%2C5347%2C2886%2C0_0&invitationId=inv_74a81b1a-7593-49c6-8fb3-82a09e64312e

6. Normalization & Schema

1. Normalization Strategy (3NF)

- Each table stores a single type of entity, ensuring clarity and modular design
- All non-key attributes depend only on the primary key
- Eliminates redundancy and avoids duplicated data

2. Handling Complex Relationships

- Many-to-many relationships are resolved through bridge tables, including:
 - listing_ownership
 - transaction_client_role
 - open_house_attendance
- Ensures flexibility in representing real-world interactions

3. Benefits of the Design

- Prevents update, insert, and delete anomalies
- Maintains data integrity and consistency across tables
- Enables efficient SQL queries using joins and aggregations

4. Analytical Advantage

- Clean schema structure supports:
 - Cross-table analysis
 - Complex business queries
 - Scalable reporting and dashboards

7. ETL Process

Extract	Transform	Load	Key Outcome
➤ Dataset is generated using AI based on business rules and schema design ➤ Data is produced in structured CSV format aligned with real estate operations	➤ Standardized formats for dates and timestamps ➤ Ensured alignment with schema definitions ➤ Validated foreign key relationships and data consistency ➤ Prepared data to support normalized, listing-centered structure	➤ Data loaded into PostgreSQL using Python scripts ➤ Tables inserted in dependency order to maintain referential integrity ➤ Post-load validation performed using row count checks	➤ Clean, structured, and analysis-ready dataset ➤ Minimal data inconsistency due to schema-aligned generation ➤ Supports both: ● Operational tracking ● Advanced analytical workflows

8. Key Analytics

Overview

- We developed **10 analytical procedures** to support business decision-making
- These are grouped into **three key categories**: performance, market, and customer behavior

1. Performance & Conversion Analytics

Metrics

- Funnel Conversion Rate
- Commission & Revenue Analysis
- Repeat Listing Detection

Insights

- Significant variation in agent performance and conversion rates
- Top agents contribute disproportionately to total revenue
- Repeated listings often indicate pricing or demand issues

Business Value: Supports agent evaluation, improves conversion efficiency, and identifies underperforming listings

2. Market & Demand Analytics

Metrics

- Average Days on Market
- Neighborhood Demand Index
- Rental vs Sale Revenue Mix

Insights

- Differences in liquidity across neighborhoods
- High-demand areas show stronger inquiry activity
- Sales dominate revenue, while rentals provide stable income

Business Value: Guides pricing strategy, market focus, and seasonal planning

3. Client Behavior Analytics

Metrics

- Client Engagement Score
- Inquiry Channel Effectiveness
- Preference Match Rate
- Open House Effectiveness

Insights

- High engagement clients are more likely to convert
- Website channel shows strongest conversion performance
- Matching accuracy can be improved to reduce inefficiency

Business Value: Enables lead prioritization, marketing optimization, and better client targeting

9. Dashboards (Metabase)

1. Dashboard Capabilities

- **Executive Revenue Overview**
 - total commission, transaction volume, revenue trends
- **Agent & Office Performance**
 - top agents, deal counts, conversion rates
- **Client Behavior Analysis**
 - engagement score, channel effectiveness, preference match rate
- **Market Performance**
 - days on market, neighborhood demand index, open house effectiveness

2. Target Users

- Executives (CEO, CFO) → strategic decision-making
- Sales managers → performance tracking and optimization
- Marketing teams → customer behavior and channel analysis

3. Business Value

- Provides real-time visibility into key KPIs
 - Eliminates need for SQL for non-technical users
 - Enables faster, data-driven decisions
- Bridges the gap between data and executive action

Dashboard Visualization:

<http://localhost:3000/public/dashboard/93105a12-a81b-451f-a99a-3ed64f420828>

10. Conclusion

1. Project Achievements

- Designed a fully normalized relational database (3NF)
- Built an end-to-end pipeline:
 - data generation → ETL → PostgreSQL → dashboards
- Developed analytical procedures to support business insights and decision-making

2.	Key Insights Enabled
➤	Listing conversion performance across agents
➤	Differences in agent productivity and commission generation
➤	Client engagement patterns across multiple interaction channels
➤	Market demand variations across neighborhoods and property types

3. Business Value of the System

- Improves data integrity and consistency
- Enables cross-process analysis using SQL joins and aggregations
- Supports both:
 - Technical analysis (analysts)
 - Dashboard-based decision-making (executives)

4. Future Opportunities

- Automating data ingestion and ETL pipelines
- Developing predictive models (e.g., demand forecasting, conversion prediction)
- Building recommendation systems for better client-property matching
- Expanding dashboards with advanced filters and real-time updates



Thank you!